Bilinear Pooling for Paired Egocentric Interaction Recognition

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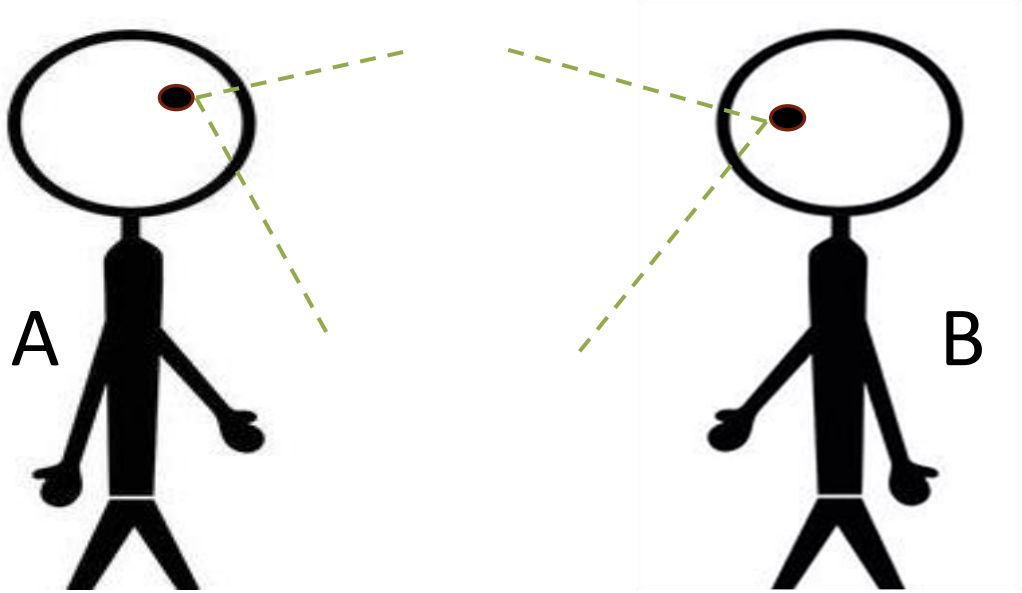
Paired Egocentric Interaction Recognition

- What is Paired Egocentric Interaction Recognition (PEIR)？
  - Input:

    Two egocentric videos taken by wearable cameras from two views of double persons face to face.
  - Output: Interaction happened between two persons.



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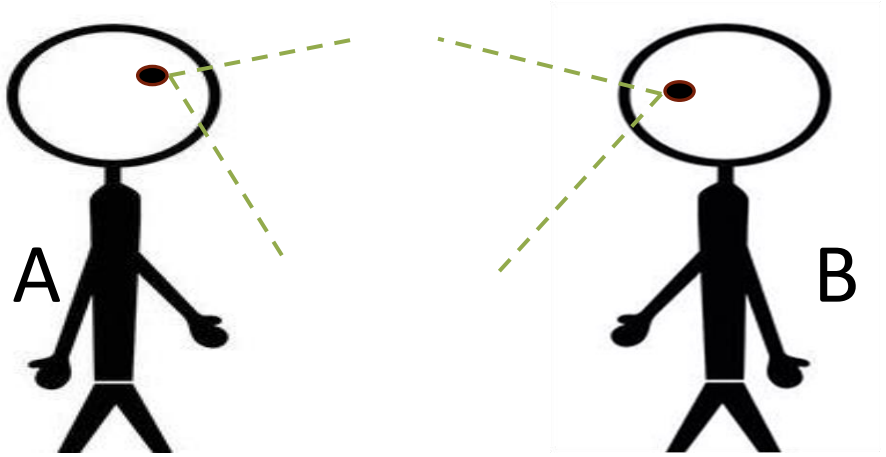
From EAR to PEIR

- Egocentric Action Recognition(EAR)
  - Input: Single egocentric video
  - Lack information of camera wearer
- PEIR
  - Previous work: Ryo Yonetani (in CVPR 2016) proposed the PEIR.
  - Application: improve AR system or a training system for competitive sports etc.



Paired Egocentric Interaction Recognition

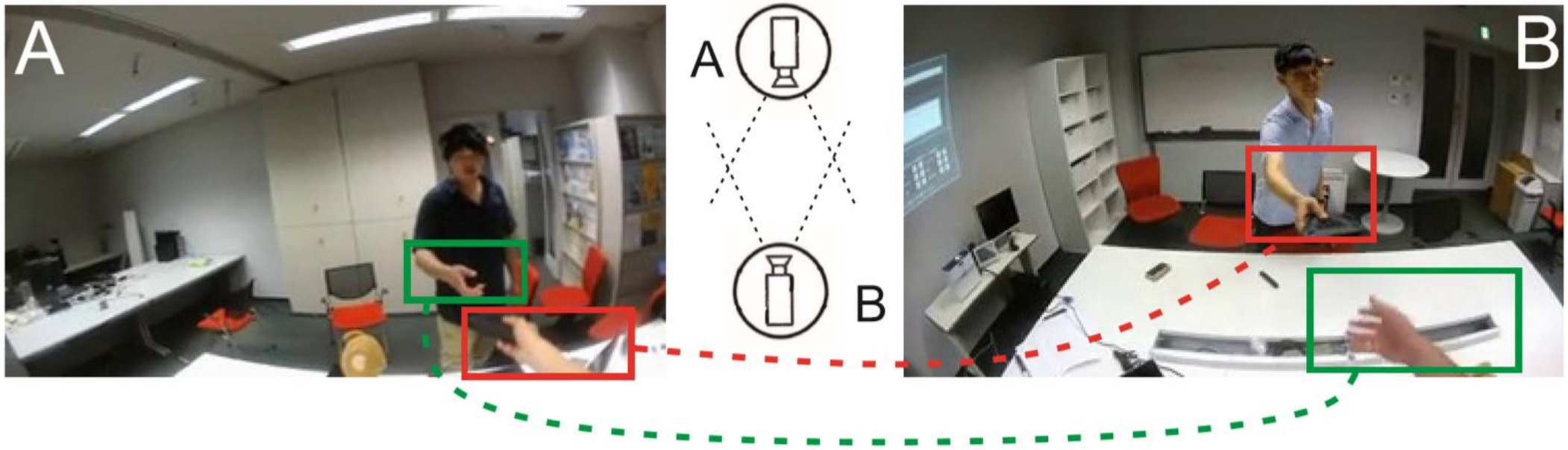
A is the one start the interaction, B is another one involving

Giving

Negative

Motivation of our approach

- Previous work ignore the consistency of interaction between two views including:
  - Common recorded parts
  - Head motion



Framework for PEIR

$$\mathbf{f}^{\mathrm{A}} = \mathrm{CNN}(v^{\mathrm{A}}),$$
$$\mathbf{f}^{\mathrm{B}} = \mathrm{CNN}(v^{\mathrm{B}}),$$
$$\mathbf{g} = \Phi(\mathbf{f}^{\mathrm{A}}, \mathbf{f}^{\mathrm{B}}),$$
$$p(l|v^{\mathrm{A}}, v^{\mathrm{B}}) = \sigma(\mathbf{W}^{\top}\mathbf{g} + \mathbf{b}),$$

Analysis for different choices of Ø

- Concatenation

  f_i^a and f_j^b are independent

$$p(l|v^{\mathrm{A}}, v^{\mathrm{B}}) \propto \prod_{i=1}^{2C} \exp\left(\mathbf{W}_i^\top \mathbf{g}_i\right),$$
$$= \prod_{i=1}^{C} \exp\left(\mathbf{W}_i^\top \mathbf{f}_i^{\mathrm{A}}\right) \prod_{i=C+1}^{2C} \exp\left(\mathbf{W}_i^\top \mathbf{f}_j^{\mathrm{B}}\right)$$

- Element-wise summation

  f_i^a and f_i^b are independent

$$p(l|v^{\mathrm{A}}, v^{\mathrm{B}}) \propto \prod_{i=1}^{C} \exp\left(\mathbf{W}_i^\top (\mathbf{f}_i^{\mathrm{A}} + \mathbf{f}_i^{\mathrm{B}})\right)$$
$$= \prod_{i=1}^{C} \exp\left(\mathbf{W}_i^\top \mathbf{f}_i^{\mathrm{A}}\right) \prod_{i=1}^{C} \exp\left(\mathbf{W}_i^\top \mathbf{f}_i^{\mathrm{B}}\right)$$

- Element-wise product

  f_i^a and f_i^b are relevant

  Only pair (i,i) is modeling

$$p(l|v^{\mathrm{A}}, v^{\mathrm{B}}) \propto \prod_{i=1}^{C} \exp\left(\mathbf{W}_i^\top \mathbf{f}_i^{\mathrm{A}} \mathbf{f}_i^{\mathrm{B}}\right)$$

Bilinear Pooling

- To capture the correlations in all pairs between two views:

$$\mathbf{g}_k = \phi(\mathbf{f}_i^{\mathrm{A}}, \mathbf{f}_j^{\mathrm{B}})$$

$$p(l|v^{\mathrm{A}}, v^{\mathrm{B}}) \propto \prod_{i=1,j=1}^{C} \exp\left(\mathbf{W}_i^{\top} \mathbf{f}_i^{\mathrm{A}} \mathbf{f}_j^{\mathrm{B}}\right)$$

Bilinear Pooling

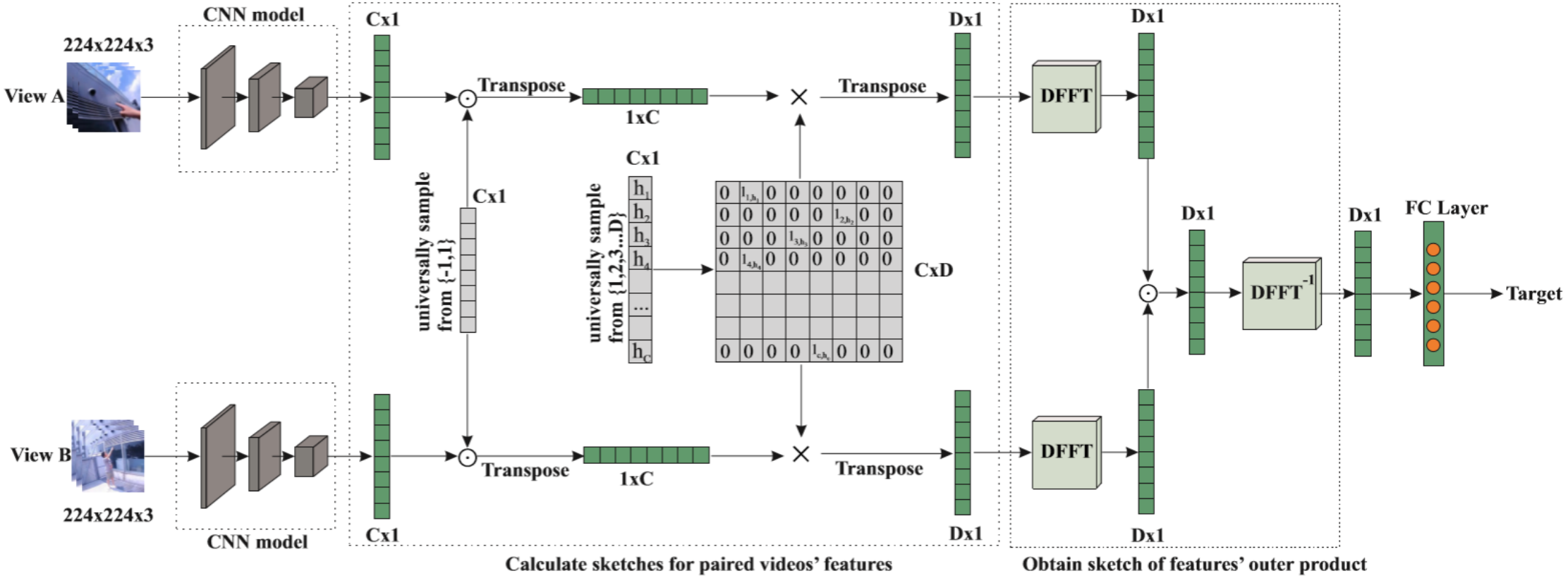
- Using Tensor Sketch of outer product $f^A \otimes f^B$ for classification.
  - In real domain

$$g = f^A \otimes f^B$$

  - In Fourier domain

$$sketch_g = sketch_{f^A} \odot sketch_{f^B}$$

Model Design



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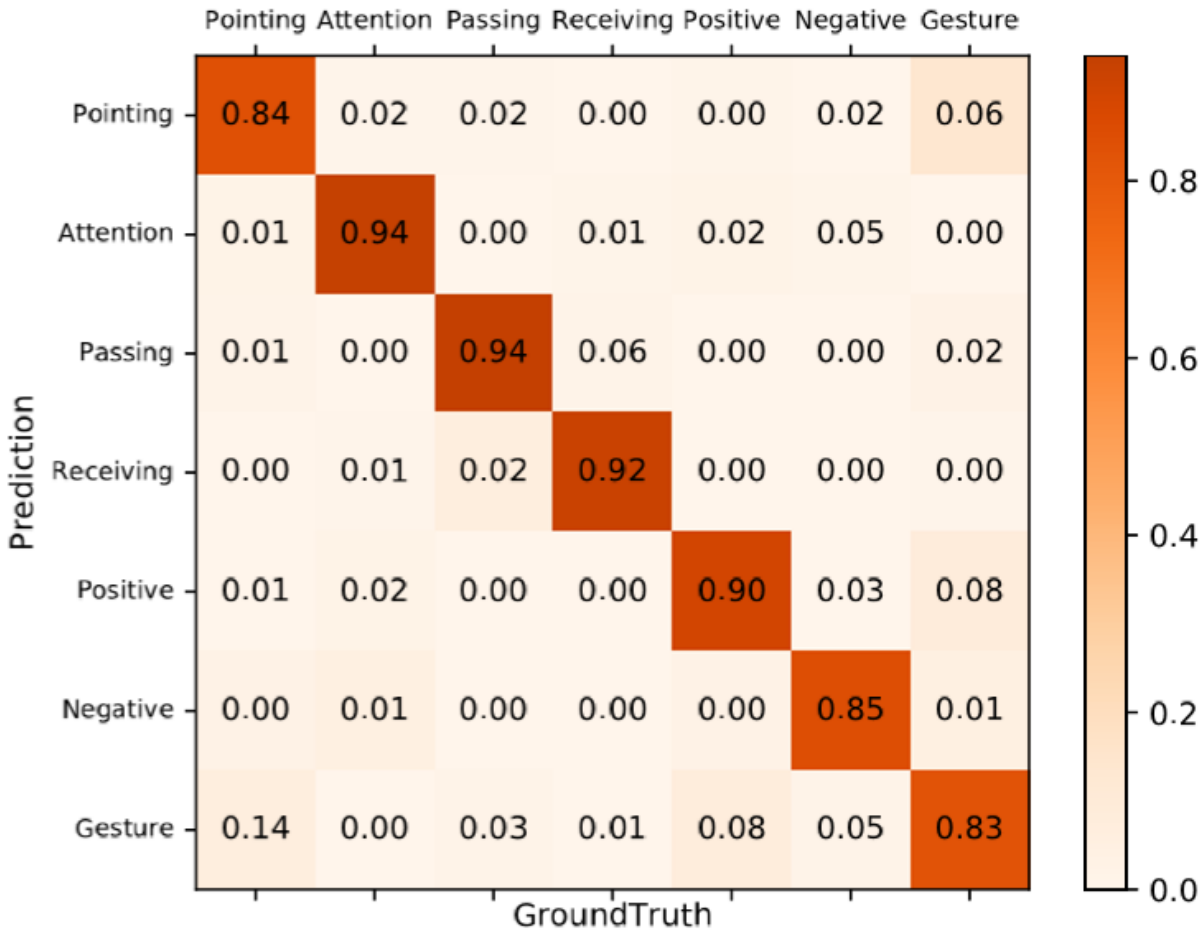
Experiments

- Paired Egocentric Video Dataset (Ryo … et. al. CVPR 2016)
  - Label: 7 kinds of interaction defined in this dataset including Pointing, Attention, Passing, Receiving, Positive, Negative, Gesture.
  - Resolution: 320x180
  - Length: 90 frames
  - Counts: 1226 pairs of videos(including non-interaction videos)

Experiments: Comparison

Modality	View	Method	D	Split1	Split2	Split3	Avg Acc
RGB	A		-	78.50	77.94	76.19	77.54
RGB	B		-	73.52	68.73	71.75	71.33
RGB	A+B	Avg	-	80.50	80.55	76.80	79.28
RGB	A+B	Svm	-	**84.96**	78.70	80.40	81.35
RGB	A+B	Concat	2048	82.25	80.89	79.30	80.81
RGB	A+B	Ours	2048	84.35	**83.09**	**83.47**	**83.63**
OF	A		-	84.96	81.35	84.24	83.51
OF	B		-	79.74	78.00	78.46	78.73
OF	A+B	Avg	-	86.01	82.62	85.86	84.83
OF	A+B	Svm	-	**89.40**	83.42	87.97	86.93
OF	A+B	Concat	2048	87.37	87.56	88.21	87.71
OF	A+B	Ours	2048	88.10	**89.06**	**89.12**	**88.76**
RGB	A+B	Sum	1024	82.99	79.14	79.09	80.41
RGB	A+B	Product	1024	84.35	79.10	72.81	78.75
RGB	A+B	Ours	1024	**86.08**	**81.34**	**83.78**	**83.73**
OF	A+B	Sum	1024	87.24	87.26	87.64	87.78
OF	A+B	Product	1024	87.30	88.44	88.34	88.03
OF	A+B	Ours	1024	**89.33**	**88.81**	**88.37**	**88.84**

Experiments: Confusion Matrix



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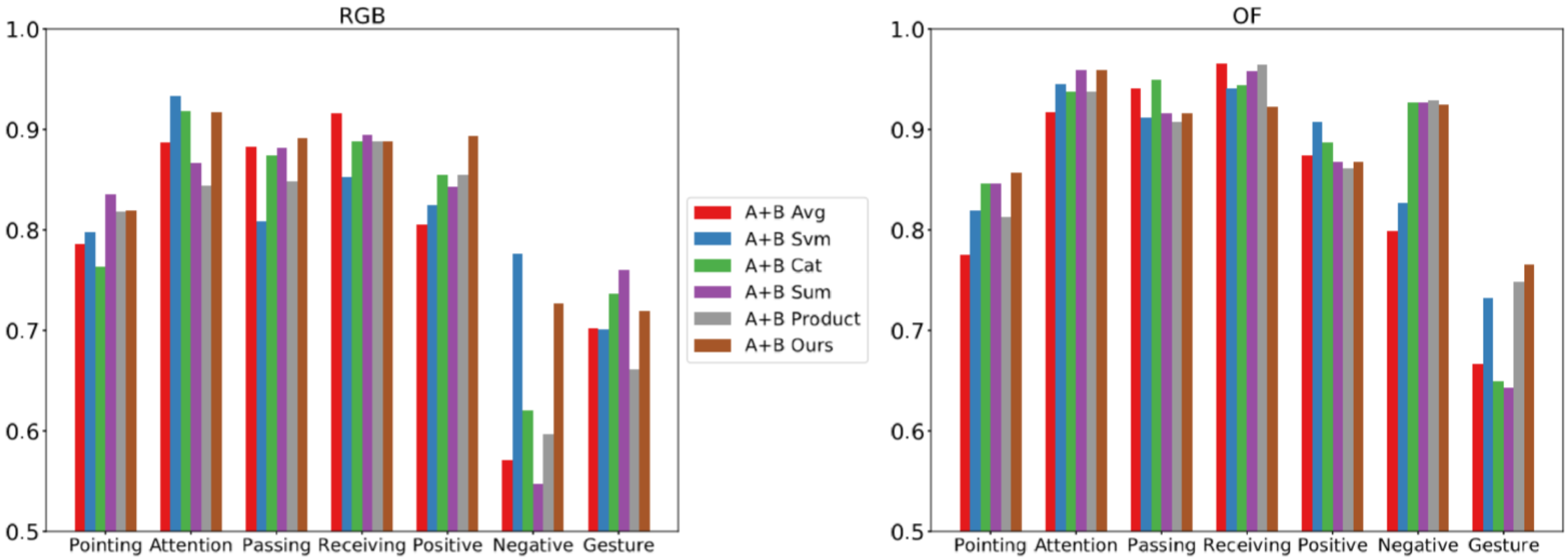
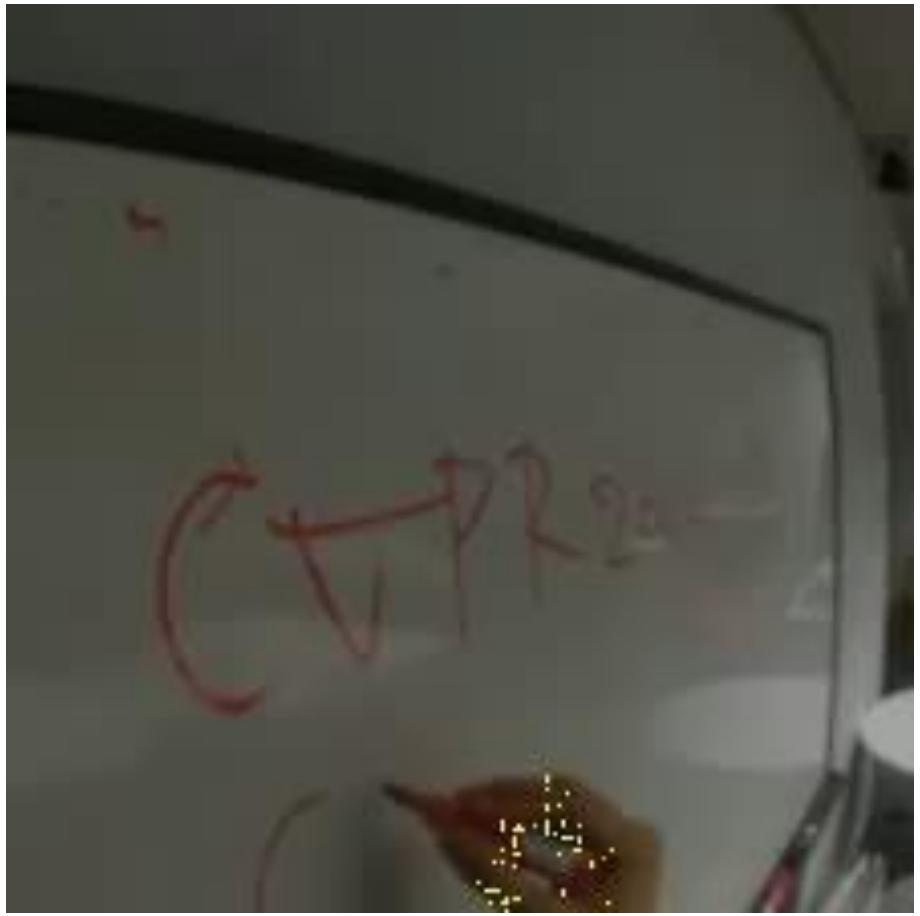
Experiments: Acc for each Category



Visualization: Integrated Gradients



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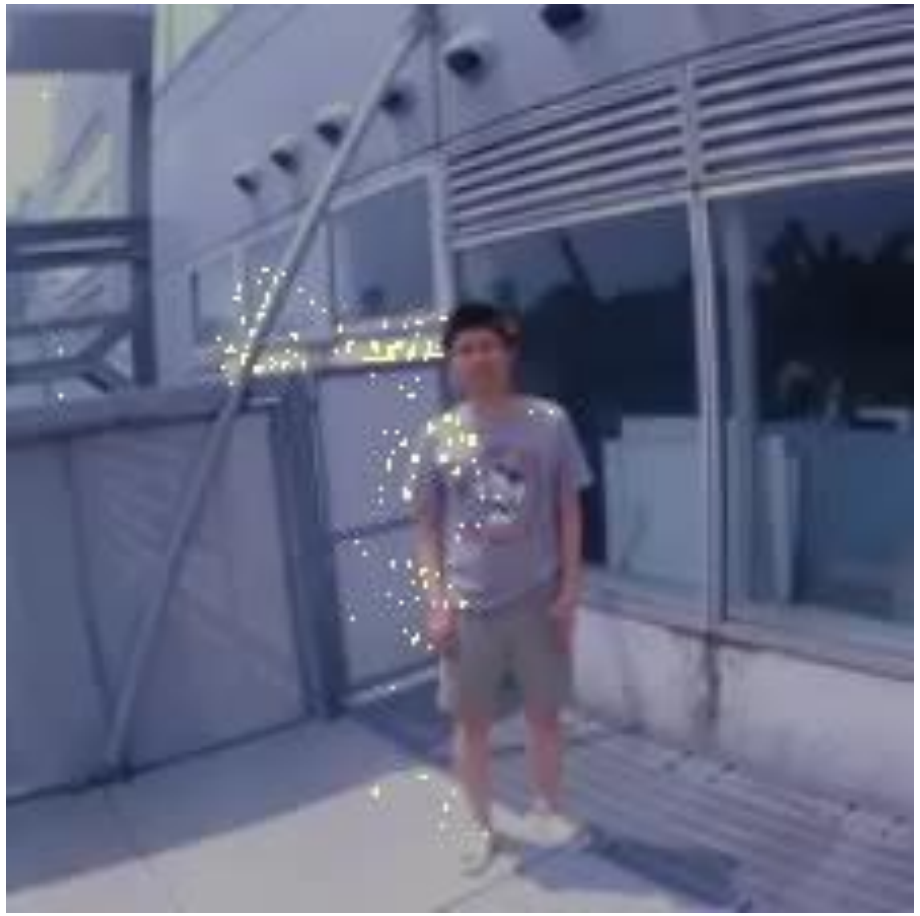

Conclusion

- The consistency of interaction exists between paired videos in PEIR task.
- Including this consistency is helpful for Paired Egocentric Interaction Recognition task.
- We achieved a new state-of-the-art performance in PEV dataset.

QA

Thanks for your listening !

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